**Thesis**
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# 1 Literature Review

Employer learning researches examine how firms update their assessments of workers as employment reveals information about productivity. Spence (1973) examines education as a way to signal ability when employers cannot observe true ability directly. Besides, Farber and Gibbons (1996) pay attention to wage dynamics, where measures of unobserved productivity should become more strongly associated with wages as experience accumulates. Further, Altonji and Pierret (2001) develop a related test using the NLSY79. Using wage regressions containing schooling and productivity proxy, they find that the association with the proxy strengthens while the schooling coefficient declines with experience. Though these papers differ in regression specifications and predictions for schooling, they share the same idea on information acquisition.

[Question on how to write a literature review]

# 2 Theoretical Model Setup

Let $i$ index workers, $t \in \mathcal{T}_i$ be the observed year, $k_{it}$ be the employer number, $j_{it} \in \{1, \ldots, J\}$ be the occupation, and $\tau < t$ represents earlier performance observations. Let $e_{it}$ denote cumulative experience and $x_{it}$ denote tenure with the current employer. Subscripts $C$ and $O$ represent the incumbent and the outside employer.

## 2.1 Two period Model

Altonji and Pierret (2001) specifies log productivity as a function of schooling $s_i$, employer observed information $q_i$,researcher observed ability estimation $z_i$, and remaining unobserved productivity $\eta_i$. Zhang instead starts from productivity in levels, $\chi_{if} = \eta_i + \delta_{if}$, where matching quality $\delta_{if}$ is known before production.

We formulate the equation as

$$p_{ijt}^k = a_j + r_j s_i + \alpha_{1j} q_i + \Lambda_j z_i + \eta_{ij} + \delta_i^k + H_j(e_{it}, x_{it})\,. \tag{1}$$

Here we assume $p$ is the log productivity, so the productivity level is $Y = \exp(p)$. Following Zhang's assumption, the match component is know to the corresponding employer and is independently recalculated for new employers. The function $H_j(e, x)$ allows experience and tenure to affect productivity independently of employer learning.

Following Altonji and Pierret, assume

$$z_i = \gamma_1 q_i + \gamma_2 s_i + \nu_i \,, \quad \eta_{ij} = \alpha_{2j} s_i + e_{ij} \,, \tag{2}$$

where $\mathrm{E}[\nu_i \mid s_i, q_i] = 0$ and $\mathrm{E}[e_{ij} \mid s_i, q_i] = 0$.

Substituting eq. (2) into eq. (1) gives

$$\begin{aligned} p_{ijt}^k =& a_j + r_j s_i + \alpha_{1j} q_i + \Lambda_j \left(\gamma_1 q_i + \gamma_2 s_i + \nu_i\right) \\ &+ \alpha_{2j} s_i + e_{ij} + \delta_i^k + H_j(e_{it}, x_{it}) \,. \end{aligned}$$

Combining the predictable and initially unpredictable terms gives

$$p_{ijt}^k = B_{ij} + u_{ij} + \delta_i^k + H_j(e_{it}, x_{it}) \,, \tag{3}$$

where

$$B_{ij} = a_j + (r_j + \Lambda_j \gamma_2 + \alpha_{2j}) s_i + (\alpha_{1j} + \Lambda_j \gamma_1) q_i \,,$$

and

$$u_{ij} = \Lambda_j \nu_i + e_{ij} \,.$$

$B_{ij}$ is the employer's initial assessment of log productivity in occupation $j$, and $u_{ij}$ is its initial prediction error. Assume $u_{ij} \mid s_i, q_i \sim \mathcal{N}\left(0, \sigma_{0j}^2\right)$.

Altonji and Pierret construct a residual performance signal by subtracting employers' initial productivity assessment from observed noisy performance. Zhang's modification is that the performance signal is observed only by the incumbent.

During job $k$, let the employer observe

$$\xi_{i\tau}^k = p_{ij\tau}^k + \varepsilon_{i\tau}^k \,, \quad \varepsilon_{i\tau}^k \sim \mathcal{N}\left(0, \sigma_{\nu j}^2\right) \,,$$

with independent noise across observations and independent of persistent productivity.

Using eq. (3), we can subtract the known components

$$\begin{aligned} d_{i\tau}^k &= \xi_{i\tau}^k - B_{ij} - \delta_i^k - H_j(e_{i\tau}, x_{i\tau}) \\ &= u_{ij} + \varepsilon_{i\tau}^k \,, \end{aligned} \tag{4}$$

and define the employer's own performance record as

$$D_{it}^k = \left\{ d_{i\tau}^k : \tau < t, \tau \text{ belongs to job } k \right\} .$$

**Deriving Zhang's learning weight**

Suppose the first employer has observed $n$ signals before the first mobility decision. From eq. (4),

$$\bar{d}_{it}^1 = u_{ij} + \bar{\varepsilon}_{it}^1 \,, \quad \mathrm{Var}(\bar{\varepsilon}_{it}^1) = \frac{\sigma_{\nu j}^2}{n} \,.$$

Therefore, $\mathrm{Cov}(u_{ij}, \bar{d}_{it}^1) = \sigma_{0j}^2$, and $\mathrm{Var}(\bar{d}_{it}^1) = \sigma_{0j}^2 + \frac{\sigma_{\nu j}^2}{n}$. As these variables are jointly normal, the conditional mean is

$$\begin{aligned} \mathrm{E}[u_{ij} \mid s_i, q_i, D_{it}^1] &= \frac{\mathrm{Cov}(u_{ij}, \bar{d}_{it}^1)}{\mathrm{Var}(\bar{d}_{it}^1)} \, \bar{d}_{it}^1 \\ &= \frac{\sigma_{0j}^2}{\sigma_{0j}^2 + \sigma_{\nu j}^2/n} \, \bar{d}_{it}^1 \,. \end{aligned}$$

Hence,

$$\hat{u}_{ijt}^{C,1} = \theta_j(n) \bar{d}_{it}^1, \quad \theta_j(n) = \frac{n\sigma_{0j}^2}{\sigma_{\nu j}^2 + n\sigma_{0j}^2} \,. \tag{5}$$

The corresponding posterior variance is

$$V_j(n) = \left( \frac{1}{\sigma_{0j}^2} + \frac{n}{\sigma_{\nu j}^2} \right)^{-1} = \frac{\sigma_{0j}^2 \cdot \sigma_{\nu j}^2}{\sigma_{\nu j}^2 + n\sigma_{0j}^2} \,,$$

where $n$ represents the performance observations, which equals tenure measured in each job periods. Specifically, for $n = 1$, eq. (5) matches with Zhang's eq. (1).

Taking expectations in eq. (3) and then using eq. (5),

$$\mathrm{E}[p_{ijt}^1 \mid \mathcal{I}_{it}^C] = B_{ij} + \delta_i^1 + H_j(e_{it}, x_{it}) + \theta_j(n) \bar{d}_{it}^1 \,. \tag{6}$$

Substitute

$$\bar{d}^1_{it} = \Lambda_j(z_i - \gamma_1 q_i - \gamma_2 s_i) + e_{ij} + \bar{\varepsilon}^1_{it}$$

into eq. (6), we get

$$\begin{aligned}\mathrm{E}[p^1_{ijt} \mid \mathcal{I}^C_{it}] =& a_j + [r_j + \alpha_{2j} + (1 - \theta_j(n))\Lambda_j\gamma_2]\, s_i \\ &+ [\alpha_{1j} + (1 - \theta_j(n))\Lambda_j\gamma_1]\, q_i \\ &+ \theta_j(n)\Lambda_j z_i + \theta_j(n)e_{ij} + \theta_j(n)\bar{\varepsilon}^1_{it} + \delta^1_i + H_j(e_{it}, x_{it})\,.\end{aligned}$$

For $F \in \{C, O\}$, define

$$P^F_{it} = \mathrm{E}[Y^F_{it} \mid \mathcal{I}^F_{it}] = \exp[R_{ij^F} + \delta^F_i + H_{j^F}(e_{it}, x^F_{it})]\mathrm{E}[\exp(u_{ij^F}) \mid \mathcal{I}^F_{it}]\,, \tag{7}$$

where $j^C$ is the incumbent occupation, $j^O$ the offered occupation, and $x^O_{it} = 0$ for a new match. For the first incumbent's normal posterior, eq. (5) and eq. (7) imply x

$$P^O_{it_1} = \exp\left[B_{ij^O} + \delta^O_i + H_{j^O}(e_{it_1}, 0) + \frac{1}{2}\sigma^2_{0j^O}\right]\,.$$

**Endogenous mobility and the informational content of employment history**

Zhang assumes that the outsider offers first, then the incumbent counteroffers, and finally the worker compares wages plus non-pecuniary job preferences. Let

$$\theta^C_{it}, \theta^O_{it} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_\theta)\,, \quad s_\theta = \sqrt{2}\sigma_\theta\,.$$

The worker moves when

$$W^O_{it} + \theta^O_{it} \geq W^C_{it} + \theta^C_{it}\,.$$

Since the taste difference is $\mathcal{N}(0, 2\sigma^2_\theta)$,

$$\Pr(M_{it} = 1 \mid W^C_{it}, W^O_{it}) = \Phi(A_{it})\,, \quad A_{it} = \frac{W^O_{it} - W^C_{it}}{s_\theta}\,. \tag{8}$$

From $\chi_c = \eta + \delta_c$ where $n \sim \mathcal{N}(\mu_n, \sigma_\eta^2)$, and $y_{c1} = \chi_c + \varepsilon_{c1}$ where $\varepsilon_{c1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, define

$$\begin{aligned} P^C &\equiv \mathrm{E}[\chi_c \mid y_{c1}, \delta_c] = \mu_n + \delta_c + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}[y_{c1} - (\mu_\eta + \delta_c)] \\ &= \left(1 - \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\right)(\mu_\eta + \delta_c) + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\, y_{c1} \\ &= \frac{\sigma_\varepsilon^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}(\mu_\eta + \delta_c) + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\, y_{c1}\,. \end{aligned}$$

As the outside employer does not observe $y_{c1}$, in the two period specification, its assessment will rely on the population mean of ability and its own known match quality. Therefore, we can write

$$P^O \equiv \mathrm{E}[\eta + \delta_o \mid \delta_o] = \mu_\eta + \delta_o\,.$$

Then following Zhang's eq. (2) and (3), we can construct optimization problem in the format of

$$\max_{W^C}\left(P^C - W^C\right)(1 - \Phi(A)) \quad \text{and} \quad \max_{W^O}(P^O - W^O)\Phi(A)\,.$$

Taking first order derivative to $W^C$ and $W^O$, respectively, gives

$$\begin{cases} W^C = & P^C - s_\theta \frac{(1-\Phi(A))}{\phi(A)}\,, \\ W^O = & P^O - s_\theta \frac{\Phi(A)}{\phi(A)}\,. \end{cases}$$

As $((1 - \Phi(A))/\phi(A))' < 0$, holding the outside offer fixed, differentiate $W^C$ equation above gives

$$dW^C = dP^C - s_\theta \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' dA\,. \tag{9}$$

Holding the outside offer $W^O$ and $s_\theta$ fixed,

$$dA = -\frac{dW^C}{s_\theta}\,.$$

Substituting this into (9) gives

$$dW^C = dP^C - s_\theta \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' \left(-\frac{dW^C}{s_\theta}\right) = dP^C + \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' dW^C\,.$$

Using the fact that $\Phi'(A) = \phi(A)$ and $\phi(A) = 1/\sqrt{2\pi} \cdot e^{-A^2/2} \Rightarrow \phi'(A) = -A\phi(A)$,

$$\left.\frac{\partial W^C}{\partial P^C}\right|_{W^O, s_\theta} = \frac{1}{1 - \left(\frac{1-\Phi(A)}{\phi(A)}\right)'} = \frac{1}{2 - A\left(\frac{1-\Phi(A)}{\phi(A)}\right)}. \tag{10}$$

As

$$\left(\frac{1-\Phi(A)}{\phi(A)}\right)' = -1 + A\left(\frac{1-\Phi(A)}{\phi(A)}\right) < 0\,,$$

We have

$$0 < \frac{\partial W^C}{\partial P^C} < 1\,,$$

which indicates that holding outside offer and taste dispersion fixed, an increase in the incumbent's assessment of worker productivity raises its wage offer by less than the productivity increase. The positive response reflects the employer's stronger incentive to

an increase in the incumbent's expected productivity raises its wage offer, but retain a worker it now considers more productive. However, the worker receives only part of the additional expected productivity through higher wages; the remainder increases the employer's marginal profit conditional on retention. Thus, favorable information acquired through employer learning raises wages, but its effect is only partially transmitted to the worker. The implications for wage growth with tenure or experience depend on how these variables affect productivity assessments and outside offers.

# 3 Data, Sample Construction, and Measurement

## 3.1 Datasets

The empirical analysis uses the National Longitudinal Survey of Youth 1979 (NLSY79) and the National Longitudinal Survey of Youth 1997 (NLSY97). Both surveys provide longitudinal information on wages, employment histories, occupations, education, demographics, and ASVAB test scores, allowing workers to be followed across employers and over labor-market experience. Due to difference in the employment history structure, the two cohorts are cleaned separately. In both datasets, negative survey codes are treated as missing values. Hourly wages are converted from cents to dollars and adjusted to 2023 real dollars using the CPI.

For the NLSY79 dataset, employment durations, which measured in years, are ordered

chronologically within each job period of the worker and are accumulated to construct actual labor market experience, with having the reported within job gaps subtracted from employment durations. Education is fixed at the value observed at the beginning of the worker's first job so that schooling does not change with subsequent labor market experience increase. For the NLSY97 dataset, employers are identified using employer UIDs. The main employer is first identified in each survey year. When it changes between adjacent surveys, the employment roster is used to distinguish the previous and new employers. Employer spells are then numbered sequentially within worker according to the sequence of employer UID. The main analysis will focus on the first three employer spells, while workers are not required to be observed in all three jobs.

## 3.2 Dependent and Independent Variables

[Adding citations of reasons for using asvab not afqt]

The dependent variable is the log real hourly wage, denote by $\ln w_{it}$, where $w_{it}$ is the CPI adjusted hourly wage of worker $i$ in year $t$.

The main independent variables are schooling, ability, labor-market experience, employer tenure, occupation, and prior mobility. Schooling is measured using education observed at the beginning of the first employer spell and is then held fixed over the worker's subsequent employment history.

Worker ability is constructed from ASVAB subtest scores. In the NLSY79, individual ASVAB components are standardized within birth-year-by-quarter groups, after which principal-components analysis is applied. The first principal component is used as the ability measure and explains approximately $68.6\%$ of the variation in the standardized scores.The NLSY97 ASVAB variables are first reconstructed by combining the positive and negative ability estimates for eight subtests: General Science, Arithmetic Reasoning, Word Knowledge, Paragraph Comprehension, Numerical Operations, Coding Speed, Mathematics Knowledge, and Mechanical Comprehension. The subtests are then standardized using the same birth-year-by-quarter procedure and summarized using the first principal component, which explains approximately $67.2\%$ of their standardized variation.

For both cohorts, actual experience is calculated from cumulative weeks worked after subtracting within-job gaps. Employer tenure measures duration with the current employer. For both cohorts, employer spells are ordered chronologically, and prior mobility can be summarized by the number of previous employer changes.

Following grouping method used by Gibbons et al. (2005), we mapped the occupation codes into seven major groups: Professional, Managers, Sales, Clerical, Craftsmen, Operatives and Laborers, and Service, under the 1970 census occupation grouping method. For observations reported using the 2000/2002 Census occupation classification, we examine the occupational definitions of each code and assign them to the 1970 group that most closely corresponds to its occupational content.

### 3.3 Summary Statistics

Table 1
Summary Statistics

| Variable | Observations | Mean | SD | Minimum | Maxmium |
|---|---|---|---|---|---|
| Panel A: Observation-level characteristics | | | | | |
| Experience | 71,367 | 5.700 | 7.452 | 0 | 65.577 |
| Hours | 71,367 | 35.051 | 12.733 | 0 | 168 |
| Occupation category | 71,367 | 3.980 | 2.218 | 1 | 7 |
| Tenure | 71,367 | 3.884 | 6.133 | 0 | 45.192 |
| Wage | 71,367 | 18.082 | 12.257 | 2.551 | 108.119 |
| Cohort indicator | 71,367 | 0.160 | 0.367 | 0 | 1 |
| Age | 71,367 | 25.232 | 7.893 | 15 | 65 |
| Panel B: Individual characteristics | | | | | |
| ASVAB score | 13,961 | 0.017 | 2.331 | -7.300 | 7.324 |
| Education | 13,961 | 11.267 | 1.819 | 0 | 20 |
| Sex category | 13,961 | 1.496 | 0.500 | 1 | 2 |
| Race category | 13,961 | 4.867 | 2.187 | 1 | 7 |

Note. Panel A reports statistics across observations. Panel B reports one record per person, using the first nonmissing value of each listed characteristic. Occupation, sex, and race are numeric category codes. Statistics are rounded to three decimal places.

## 4 Empirical Specifications

### 4.1 OLS Model

We begin by examining how the relationship between wages, schooling, and measured ability evolves over workers' first three job spells. The goal of this section is to establish whether wages become more closely associated with ability as employers learn information about productivity, and whether this relationship varies with employment history. We distinguish cumulative labor market experience from tenure with the current employer and estimate the specification separately from the NLSY79 and NLSY97 cohorts.

The specification comes from three articles. Altonji and Pierret (2001) motivate examining whether wages become more closely associated with initially unobserved ability as employers learn, while reliance on schooling as a productivity signal declines. Gibbons et al. (2005) emphasize allowing wage coefficients to vary across occupations, reflecting differences in the valuation of worker skills. Zhang (2009) distinguishes cumulative experience from employer tenure, where incumbent employers observe performance within the employment, whereas outside employers can infer productivity from employment histories. The resulting equation is a empirical specifications constructed by these mechanisms.

Let $i$ index workers, $t$ represents calendar years, and $c \in \{79, 97\}$ indicates survey cohorts. Let $k = k_{it} \in \{1, 2, 3\}$ represents the worker's current job spell, $j = j_{it} \in \{1, \ldots, J\}$ represents the current occupation, and $D_{ijtc}$ indicate employment in occupation $j$.

Before allowing wage relationships to vary across occupations, we estimate a baseline equation separately for each cohort. Following the employer learning motivation of Altonji and Pierret (2001), we allow the associations of schooling and measured ability with wages to change with experience. We extend this specification by including interactions with employer tenure:

$$\begin{aligned}\log w_{itc} =& \alpha_{kc} + \beta_{Sc} S_{ikc} + \beta_{Zc} Z_{ic} + \gamma_{Sc} \left(S_{ikc} E_{itc}\right) + \gamma_{Zc} \left(Z_{ic} E_{itc}\right) \\ &+ \theta_{Sc} \left(S_{ikc} T_{itc}\right) + \theta_{Zc} \left(Z_{ic} T_{itc}\right) + f_c \left(E_{itc}\right) + g_c \left(T_{itc}\right) \\ &+ \lambda_{tc} + X_{itc} \eta_c + \varepsilon_{itc} \,. \end{aligned} \tag{11}$$

Here, $w_{itc}$ is the real hourly wage, $S_{ikc}$ is schooling measured at the beginning of spell $k$ and held fixed within that job spell, and $Z_{ic}$ is the standardized ASVAB based ability measure, which is observed by the researcher but assumed not to be directly observed by employers. Actual experience, $E_{itc}$, accumulates across jobs, and tenure, $T_{itc}$, measures employment with the current employer, where both of these two variables are measured in years. Occupation indicators refer to the occupation reported at survey time each year.

The functions $f_c$ and $g_c$ are cubic polynomials in the form of $x + x^2 + x^3$. The function $h_{jc}$ contains occupation specific deviations in the same experience and tenure polynomial terms. Therefore, occupational differences in ordinary wage growth are accommodated alongside differences in the schooling and ability gradients. The specification includes job spell effects $\alpha_{kc}$, calendar year effects $\lambda_{tc}$, additional demographical controls $X_{itc}$, and standard errors $\varepsilon_{itc}$ which clustered by worker.

The implied schooling and ability wage gradients are

$$b_{Sc}(E,T) = \beta_{Sc} + \gamma_{Sc}E + \theta_{Sc}T\,,$$
$$b_{Zc}(E,T) = \beta_{Zc} + \gamma_{Zc}E + \theta_{Zc}T\,.$$

Under the informational assumptions of the employer learning model, an increasing ability gradient and a decreasing schooling gradient are consistent with employers placing greater weight on observed productivity and less weight on schooling as a signal. At fixed tenure, these patterns correspond to $\gamma_{Zc} > 0$ and $\gamma_{Sc} < 0$. During the continuous employment with the same employer, experience and tenure both increase. The corresponding changes in the gradients are therefore $\gamma_{Zc} + \theta_{Zc}$ and $\gamma_{Sc} + \theta_{Sc}$. These estimates provide descriptive evidence on employer learning, as skill accumulation and worker sorting can also change wage gradients.

Introducing occupational categories, we then estimate

$$\begin{aligned}\log w_{itc} =& \alpha_{kc} + \beta_{Sc}S_{ikc} + \beta_{Zc}Z_{ic} + \gamma_{Sc}S_{ikc}E_{itc} + \gamma_{Zc}Z_{ic}E_{itc} \\ &+ \theta_{Sc}S_{ikc}T_{itc} + \theta_{Zc}Z_{ic}T_{itc} + f_c\left(E_{itc}\right) + g_c\left(T_{itc}\right) \\ &+ \sum_{j=2}^{J} D_{ijtc}\big[\delta_{jc} + \rho_{Sjc}S_{ikc} + \rho_{Zjc}Z_{ic} + h_{jc}\left(E_{itc},T_{itc}\right) + \phi_{Sjc}\left(S_{ikc}E_{itc}\right) \\ &+ \phi_{Zjc}\left(Z_{ic}E_{itc}\right) + \psi_{Sjc}\left(S_{ikc}T_{itc}\right) + \psi_{Zjc}\left(Z_{ic}T_{itc}\right)\big] + \lambda_{tc} + X_{itc}\delta_c + \varepsilon_{itc}\,. \qquad (12)\end{aligned}$$

Occupation $1$ is the reference category. For $q \in \{S, Z\}$, the conditional wage coefficient in occupation $j \neq 1$ is

$$b_{qjc}(E,T) = \beta_{qc} + \rho_{qjc} + \left(\gamma_{qc} + \phi_{qjc}\right)E + \left(\theta_{qc} + \psi_{qjc}\right)T\,, \qquad (13)$$

with the occupational deviations set to zero for $j = 1$. Consequently, $\phi_{qjc}$ and $\psi_{qjc}$ measure occupational differences in how the schooling or ability coefficient varies with experience and tenure, respectively.

During the continued employment within a given occupation, experience and tenure increase together. The corresponding change in the wage ability gradient is therefore

$$L_{Zjc} \equiv \gamma_{Zc} + \phi_{Zjc} + \theta_{Zc} + \psi_{Zjc}\,, \qquad (14)$$
$$L_{Sjc} \equiv \gamma_{Sc} + \phi_{Sjc} + \theta_{Sc} + \psi_{Sjc}\,. \qquad (15)$$

Employer learning predicts $L_{Zjc} > 0$. If schooling initially serves as a productivity signal, the analogous schooling gradient is expected to decline, $L_{Sjc} < 0$. Comparisons of these quantities across occupations and cohorts address the principal empirical questions. When experience and tenure coincide in the first spell, separating their contributions requires variation from subsequent spells.

The baseline OLS estimates allow us to compare how the wage associations of schooling and ability change with experience and tenure across occupations and cohorts. Results consistent with the predicted signs provide suggestive evidence of employer learning. However, these patterns may also reflect differences in skill development, the rewards to skills, and workers' selection into jobs. The estimates therefore cannot establish whether occupational or cohort differences arise from employer learning alone.

## 4.2 Instrument Selection

[Need thinking]

Actual labor market experience may be endogenous because employment continuity depends on worker productivity, labor supply decisions, or employer assessments. Occupational assignment and job duration may similarly respond to information about productivity. Therefore, instrument variables are helpful to distinguish endogeneity of regressors.

Altonji and Pierret (2001) instrument actual experience terms with corresponding functions of potential experience, due to the fact that realized employment may itself convey information about worker's productivity. Zhang (2009)follows this approach in his baseline and subsequently distinguishes experience from tenure when examining employer learning. Gibbons et al. (2005) address endogenous occupational sorting using instruments constructed from sector histories.

Since actual experiences are included in the model through polynomial terms and interactions with schooling, ability, and occupation. Each of these terms is treated as a separate endogenous regressor. Identifying their coefficients therefore requires at least as many linearly independent excluded instruments as endogenous regressors, together with a full-rank relationship between the instruments and these regressors conditional on the included exogenous variables.

Let $P_{itc}$ denote potential experience, measured as elapsed years since the worker's initial job market entry. This measure accumulates across the first three job spells and includes periods without employment. We adapt the potential experience approach by replacing

each actual experience term with its corresponding function of $P_{itc}$. For the common experience terms, the excluded instrument vector is

$$\left(P_{itc}, P_{itc}^2, P_{itc}^3, S_{ikc}P_{itc}, Z_{ic}P_{itc}\right) .$$

To accommodate occupational differences, we additionally interact these instruments with the non-reference occupational indicators, where the vector of instruments is

$$\left(Q_{itc}^0, D_{i2tc}Q_{itc}^0, \ldots D_{iJtc}Q_{itc}^0\right) .$$

This construction matched the polynomial order and occupational interactions in the baseline equation. Therefore we have $5 \cdot J$ excluded instruments for the $5 \cdot J$ experience related regressors in the baseline equation. Restricting the instruments to the corresponding polynomial terms and occupational interactions limits their number and the potential for first-stage overfitting.

# 5 Results

## 5.1 Baseline Model Analysis

In this subsection, we will examine how wage associates with schooling and measured ability evolve with experience and employer tenure in the NLSY79 and NLSY97 cohorts. Table 2 presents the baseline estimates with common coefficients across occupations, while Table 3 allows the coefficients to vary across occupational groups. In the analysis, we will compare coefficient magnitudes and statistical significance across cohorts, distinguish the schooling and ability gradients at zero experience and tenure from their subsequent changes. Linear combination tests assess whether these gradients change during continued employment with the same employer, while joint Wald tests assess whether the corresponding coefficients differ across occupations. We then compare the findings with the employer learning evidence of Altonji and Pierret (2001) and the analysis of skill valuation and occupational sorting in Gibbons et al. (2005), evaluating which patterns are consistent with employer learning and which will require further explanations.

Table 2
Education, Ability, and Wage Dynamics: Baseline Estimates

| | NLSY79 (1) | NLSY97 (2) |
|---|---|---|
| A. Regression coefficients | | |
| Education | $7.404^{***}$ | $3.508^{***}$ |
| | (0.365) | (0.960) |
| ASVAB | $-0.934^{***}$ | $0.344$ |
| | (0.276) | (0.714) |
| Experience | $6.752^{***}$ | $-0.273$ |
| | (1.284) | (2.723) |
| Tenure | $5.300^{***}$ | $4.788^{*}$ |
| | (1.536) | (2.689) |
| Education $\times$ Experience | $-0.171^{*}$ | $0.316$ |
| | (0.097) | (0.202) |
| ASVAB $\times$ Experience | $0.198^{***}$ | $0.132$ |
| | (0.073) | (0.190) |
| Education $\times$ Tenure | $0.229^{*}$ | $0.036$ |
| | (0.122) | (0.202) |
| ASVAB $\times$ Tenure | $0.166$ | $0.100$ |
| | (0.104) | (0.215) |
| B. Linear combination tests | | |
| Education interactions, $\gamma_S + \theta_S$ | $0.058$ | $0.353^{**}$ |
| | (0.062) | (0.174) |
| $p$ value | 0.355 | 0.043 |
| ASVAB interactions, $\gamma_Z + \theta_Z$ | $0.364^{***}$ | $0.232^{*}$ |
| | (0.057) | (0.129) |
| $p$ value | $< 0.001$ | 0.072 |
| Experience and tenure squared and cubed | Yes | Yes |
| Year and job number indicators | Yes | Yes |
| Sex and race controls | Yes | Yes |
| Observations | 59,927 | 11,440 |
| Respondents | 11,072 | 2,889 |
| $R^2$ | 0.4029 | 0.2933 |

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 3
Education, Ability, and Wage Dynamics by Occupation

| Occupation | Education | | | ASVAB | | |
|---|---|---|---|---|---|---|
| | Level (1) | × Experience (2) | × Tenure (3) | Level (4) | × Experience (5) | × Tenure (6) |
| A. NLSY79 | | | | | | |
| Clerical | 4.690*** | −0.111 | 0.112 | −1.032** | 0.330* | −0.070 |
| | ( 0.588) | ( 0.204) | ( 0.206) | ( 0.423) | ( 0.178) | ( 0.200) |
| Craftsmen | 4.908*** | 0.325 | −0.574 | 0.051 | −0.070 | 0.379** |
| | ( 1.499) | ( 0.272) | ( 0.414) | ( 0.962) | ( 0.142) | ( 0.178) |
| Managers | 5.800*** | −0.317 | 0.575 | 0.853 | 0.503 | −0.422 |
| | ( 2.049) | ( 0.325) | ( 0.350) | ( 1.399) | ( 0.336) | ( 0.356) |
| Operatives and | 5.674*** | −0.178* | 0.506** | −0.730 | 0.003 | 0.230 |
| laborers | ( 0.515) | ( 0.107) | ( 0.211) | ( 0.461) | ( 0.088) | ( 0.189) |
| Professional | 7.662*** | 0.362 | −0.584** | 1.550 | −0.175 | 0.442** |
| | ( 1.072) | ( 0.240) | ( 0.243) | ( 1.078) | ( 0.202) | ( 0.216) |
| Sales | 6.503*** | −0.806*** | 1.671*** | −2.583*** | 0.684*** | 0.074 |
| | ( 1.374) | ( 0.245) | ( 0.523) | ( 0.965) | ( 0.197) | ( 0.432) |
| Service | 7.227*** | −0.202 | 0.113 | −0.893** | 0.075 | 0.364* |
| | ( 0.518) | ( 0.147) | ( 0.170) | ( 0.352) | ( 0.096) | ( 0.190) |
| Equality test, $p$ | 0.0235 | 0.0228 | $< 0.001$ | 0.0717 | 0.0153 | 0.2179 |
| Observations: 59,927; Respondents: 11,072; $R^2 = 0.4410$. | | | | | | |
| B. NLSY97 | | | | | | |
| Clerical | 3.680** | −0.474 | 0.461 | 0.200 | 0.365 | −0.329 |
| | ( 1.517) | ( 0.441) | ( 0.381) | ( 0.865) | ( 0.225) | ( 0.298) |
| Craftsmen | 2.608 | 0.418 | −0.351 | −3.596 | −0.005 | 0.120 |
| | ( 4.758) | ( 0.864) | ( 0.844) | ( 3.732) | ( 0.829) | ( 0.752) |
| Managers | 12.092*** | 0.670 | −1.013 | 0.008 | −0.295 | 0.882 |
| | ( 2.672) | ( 0.626) | ( 0.643) | ( 3.301) | ( 0.674) | ( 0.749) |
| Operatives and | 1.121 | −0.121 | 0.371 | 0.820 | 0.224 | −0.338 |
| laborers | ( 1.417) | ( 0.258) | ( 0.298) | ( 1.216) | ( 0.312) | ( 0.352) |
| Professional | 3.559** | 0.788** | −0.383 | −2.081 | 1.013** | −0.680 |
| | ( 1.678) | ( 0.340) | ( 0.374) | ( 1.767) | ( 0.446) | ( 0.529) |
| Sales | 9.793* | −0.064 | −0.184 | −1.227 | −0.454 | 0.758 |
| | ( 5.422) | ( 0.979) | ( 1.233) | ( 1.851) | ( 0.556) | ( 0.784) |
| Service | 0.856 | 0.313 | 0.150 | 0.920 | 0.722*** | −0.518 |
| | ( 1.382) | ( 0.316) | ( 0.400) | ( 1.010) | ( 0.274) | ( 0.386) |
| Equality test, $p$ | 0.0041 | 0.2295 | 0.2977 | 0.6619 | 0.2827 | 0.4784 |
| Observations: 11,440; Respondents: 2,889; $R^2 = 0.3455$. | | | | | | |

Note. Each panel reports one unweighted OLS regression. Estimates and standard errors are expressed on the scale of 100 times log real hourly wages. Entries are total occupation coefficients. Standard errors clustered by respondent are in parentheses. Wages are winsorized at the 1st and 99th percentiles of the pooled merged dataset before restricting job numbers to 1 through 3. Education, experience, and tenure are in years. ASVAB is in principal component units. Level coefficients apply at zero experience and tenure. Both regressions include sex and race controls. Equality tests compare the corresponding coefficient across all seven occupations.
$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

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